# AN ORIENTED FLAT COUNTEREXAMPLE TO THE CENTRE-RANK FORMULA FOR TOPOLOGICAL COMPLEXITY

KAIWEN JIN

Abstract. We construct a closed connected oriented flat five-manifold whose fundamental group has centre of rank two and whose unreduced topological complexity is at least ten. Since its universal cover is Euclidean space, this gives a negative answer to the proposed formula $\mathrm{TC}(M) = 2\dim M + 1 - \operatorname{rank} Z(\pi_1 M)$ for closed oriented aspherical manifolds. The lower bound follows from a nonzero product of nine degree-one zero divisors in the mod-2 cohomology ring of a real Bott manifold.

## 1. The centre-rank question

For a path-connected space $X$, its unreduced topological complexity $\mathrm{TC}(X)$ is the least number of open subsets covering $X \times X$ on which the path endpoint map $PX \to X \times X$ admits continuous sections. Thus $\mathrm{TC}(X)$ counts the continuous local rules needed for motion planning, with both the initial and final configuration allowed to vary [1].

For a closed connected oriented aspherical $n$-manifold $M$, put

$$z = \dim_{\mathbb{Q}}(Z(\pi_1 M) \otimes_{\mathbb{Z}} \mathbb{Q}).$$

The centre-rank question asks whether

$$\mathrm{TC}(M) = 2n + 1 - z. \tag{1}$$

This formula is posed in [3, Question 5.8]. We show that it fails already among oriented flat manifolds.

**Theorem 1.** *There is a closed connected oriented flat five-manifold $M$ with contractible universal cover $\mathbb{R}^5$ such that*

$$\dim_{\mathbb{Q}}(Z(\pi_1 M) \otimes_{\mathbb{Z}} \mathbb{Q}) = 2, \qquad \mathrm{TC}(M) \geq 10.$$

*In particular, $M$ is an aspherical counterexample to (1), whose right-hand side is 9 for this manifold.*

The construction uses the affine realization and cohomology presentation of real Bott manifolds due to Kamishima and Masuda [2]. A mixed product of nine zero divisors then gives the lower bound through Farber's theorem [1, Theorem 7]. An exact value of $\mathrm{TC}(M)$ is not needed for the counterexample. Section 2 constructs the manifold and computes its centre; Section 3 gives the cup product and completes the proof. All cohomology below has coefficients in $\mathbb{F}_2$, and TC always uses the unreduced convention.

## 2. The manifold and its centre

Consider the strictly upper triangular binary matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2}$$

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For $1 \le i \le 5$, let $e_i$ be the $i$th coordinate vector and define the affine isometry

$$s_i(u) = D_i u + \tfrac{1}{2} e_i, \qquad D_i = \operatorname{diag}\big((-1)^{A_{i1}}, \dots, (-1)^{A_{i5}}\big). \tag{3}$$

Set $\Gamma = \langle s_1, \dots, s_5 \rangle$ and $M = \mathbb{R}^5/\Gamma$. These are exactly the generators in [2, equation (3.1)], and [2, Lemma 3.1] identifies this quotient with the real Bott manifold $M(A)$ and its fundamental group with $\Gamma$.

**Proposition 2.** *The manifold $M$ is closed, connected, oriented and aspherical, and* $\dim_{\mathbb{Q}}(Z(\Gamma) \otimes_{\mathbb{Z}} \mathbb{Q}) = 2$.

*Proof.* Since $A_{ii} = 0$, the square $s_i^2$ is the unit translation by $e_i$. The integer translation lattice $\mathbb{Z}^5$ is therefore a normal subgroup of $\Gamma$. On $T^5 = \mathbb{R}^5/\mathbb{Z}^5$ the five generators commute and have order two: a sign change and a half translation commute modulo integer translations. For a nonempty product of distinct generators, choose its smallest index $i$. Every selected generator of larger index fixes coordinate $i$, whereas $s_i$ translates that coordinate by $1/2$. Hence the product has no fixed point. The induced action is consequently a free action of $(\mathbb{Z}/2)^5$.

It follows that $M = T^5/(\mathbb{Z}/2)^5$ is a smooth compact connected manifold without boundary. Its Euclidean metric descends to a flat metric. The lifted action on $\mathbb{R}^5$ is free and properly discontinuous: all translation vectors lie in $\frac{1}{2}\mathbb{Z}^5$, the linear parts are diagonal sign matrices, and only finitely many such affine maps can move a compact set to meet itself. Thus $\mathbb{R}^5$ is its contractible universal cover. Every row of $A$ has even sum, so every $D_i$ has determinant $+1$ and $M$ is oriented.

Let $g(u) = Bu + v$ be central in $\Gamma$. Commuting with the five unit translations gives $Be_i = e_i$ for every $i$, so $B = I$. A translation by $v$ commutes with $s_i$ precisely when $D_i v = v$. For the matrix (2) these conditions are

$$v_2 = v_4 = v_5 = 0.$$

The unit translations in coordinates 1 and 3 are central, while every central vector lies in $\frac{1}{2}\mathbb{Z}e_1 \oplus \frac{1}{2}\mathbb{Z}e_3$. The two unit translations therefore span the centre after tensoring with $\mathbb{Q}$ and remain linearly independent under the coordinate map to $\mathbb{Q}^2$. Its rank is exactly two. □

## 3. A product of nine zero divisors

By [2, Lemma 2.1], the cohomology ring $R = H^*(M; \mathbb{F}_2)$ has degree-one generators $a, b, c, d, e$, corresponding to $x_1, \dots, x_5$, and presentation

$$R = \mathbb{F}_2[a, b, c, d, e]/(a^2,\ b^2 + ab,\ c^2,\ d^2 + (a+b+c)d,\ e^2 + (b+c)e). \tag{4}$$

Each successive relation is monic quadratic in the newly adjoined variable. Polynomial division therefore gives a free rank-two extension with basis $1, x_j$ at each stage. The squarefree monomials are a basis of $R$; in particular,

$$\alpha = abcd \in H^4(M; \mathbb{F}_2), \qquad \omega = abcde \in H^5(M; \mathbb{F}_2)$$

are nonzero.

For $x \in R$, write $\bar{x} = x \otimes 1 + 1 \otimes x$ in the graded algebra $R \otimes_{\mathbb{F}_2} R$. This lies in the kernel of cup multiplication $R \otimes R \to R$, which is the zero-divisors ideal used in [1, Definition 6]. All graded tensor signs equal 1 over $\mathbb{F}_2$.

**Lemma 3.** *In $R \otimes_{\mathbb{F}_2} R$,*

$$\bar{a}\,\bar{b}\,\bar{d}^{\,5}\,\bar{e}^{\,2} = \alpha \otimes \omega + \omega \otimes \alpha \neq 0. \tag{5}$$

*Proof.* Put $u = a + b + c$. The relations (4) give

$$u^2 = ab, \qquad u^3 = abc, \qquad u^4 = 0.$$

Indeed, $u^3 = ab(a+b+c) = abc$, since $a^2 = 0$ and $ab^2 = a^2 b = 0$. It follows from $d^2 = ud$ that

$$d^4 = u^3 d = \alpha, \qquad d^5 = u^4 d = 0.$$

The binomial coefficients $\binom{5}{j}$ are odd only for $j = 0, 1, 4, 5$. Consequently

$$\bar{d}^5 = \alpha \otimes d + d \otimes \alpha.$$

Since $a\alpha = b\alpha = 0$, multiplication by $\bar{a}\bar{b}$ leaves

$$\bar{a}\bar{b}\bar{d}^5 = \alpha \otimes abd + abd \otimes \alpha. \tag{6}$$

Finally,

$$\bar{e}^{\,2} = e^2 \otimes 1 + 1 \otimes e^2, \qquad \alpha e^2 = 0, \qquad abd\, e^2 = abd(b + c)e = \omega.$$

Multiplying (6) by $\bar{e}^{\,2}$ proves the identity. The two summands have distinct bidegrees $(4, 5)$ and $(5, 4)$ and are both nonzero, so their sum is nonzero. □

*Proof of Theorem 1.* Proposition 2 supplies the required manifold and its centre rank. The product (5) contains $1 + 1 + 5 + 2 = 9$ zero-divisor factors. Farber's lower bound [1, Theorem 7] states, in the unreduced convention, that $\mathrm{TC}(M)$ is strictly greater than the zero-divisors cup length of $H^*(M; \mathbb{F}_2)$. Hence $\mathrm{TC}(M) \geq 10$, while $2 \dim M + 1 - \dim_{\mathbb{Q}}(Z(\pi_1 M) \otimes_{\mathbb{Z}} \mathbb{Q}) = 9$. □

## AI use

OpenAI Codex was used for the finite construction search, algebraic checks, and preparation of the text. The proof is given in full through the displayed algebraic calculations and the cited cohomology and topological complexity results.

## References

[1] M. Farber, *Topological complexity of motion planning*, Discrete Comput. Geom. **29** (2003), 211–221. doi:10.1007/s00454-002-0760-9. Preprint: arXiv:math/0111197v1.

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*Email address*: yui.ui.siki@gmail.com